

Week 2 Lab Worksheet

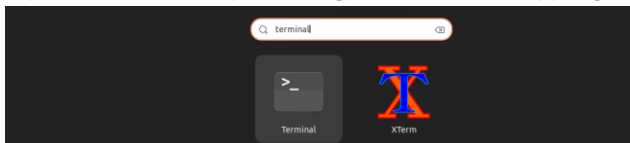
The objective of this Week 2 laboratory session was to establish a foundational understanding of network emulation and traffic analysis. By utilizing **Mininet**, we transitioned from theoretical network design to practical, programmatic topology generation (**LO2**). Furthermore, by manipulating Linux kernel networking stacks and utilizing **Wireshark**, we critically analysed the behavioural differences between Unicast, Broadcast, and Multicast transmission modes (**LO3 & LO4**), providing empirical evidence of how network traffic propagates across a virtualized switch fabric."

Learning Objectives

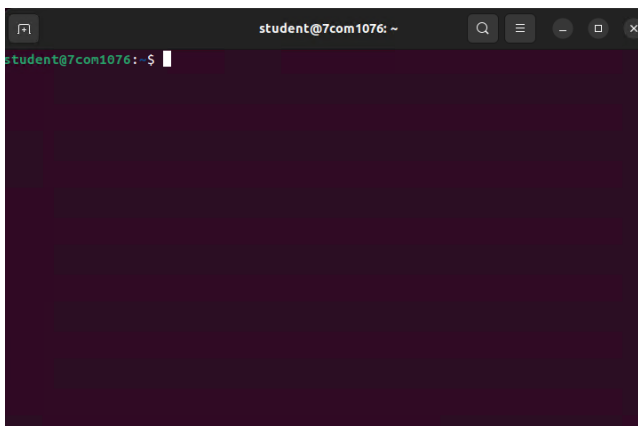
1. Deploy and navigate network emulation environments to verify virtual infrastructure.
2. Programmatically design and customize network topologies using Python-based APIs.
3. Configure and differentiate fundamental network transmission modes
4. Utilise packet analysis tools to inspect and validate traffic behaviours.

1. Introduction to Mininet

1. Open a terminal by clicking Activities and typing terminal



The following window will appear.



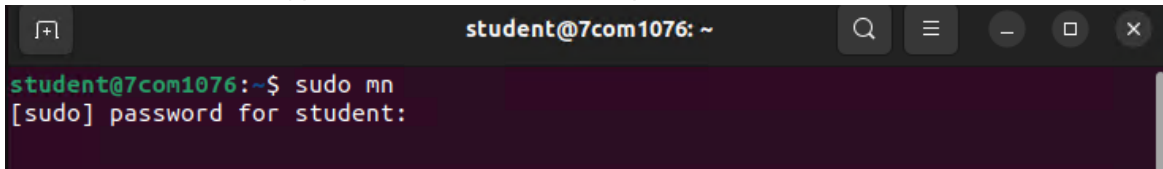
2. In the terminal, type `ip a` and press **Enter**. Record the following information from the output:

```
student@7com1076: ~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:3b:f7:08 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.170.135/24 brd 192.168.170.255 scope global dynamic noprefixroute ens33
        valid_lft 1345sec preferred_lft 1345sec
    inet6 fe80::d283:6deb:cef4:9054/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

- IP address v4 and v6
- Netmask
- Broadcast IP address v4
- MAC address

3. Launching Mininet

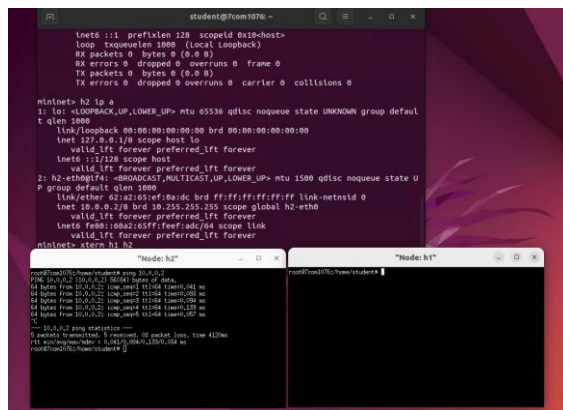
- i. In the same terminal, type `sudo mn` and enter the password 123 to start Mininet



```
student@7com1076: ~
student@7com1076:~$ sudo mn
[sudo] password for student:
```

- ii. Once in Mininet, use the following commands to explore the network topology:

- a. `h1 ip a`: Displays the configuration of host 1.
- b. `h2 ip a`: Displays the configuration of host 2.
- c. `xterm h1 h2`
- d. In the h1 terminal, type `ping 10.0.0.2` to check connectivity between hosts. The ping should succeed. Press **CTRL + C** to stop the ping.



- e. `exit`: Exit Mininet

2. Introduction to Python API

1. Navigate to 7COM2008 Folder in Files. Double click LAB1 Folder.
2. Open the provided `lab1.py` Python script and complete the following tasks: Write the following commands underneath the `#` symbol. **Ensure proper indentation as Python is sensitive to it.**
3. Under **#Add h4** type in the following command
 - a. `h4 = self.addHost('h4', ip='10.0.0.4/24')`
4. Under **#Connect h4 to s1** type in the following command
 - a. `self.addLink(h4, s1)`
5. Save by pressing **CTRL + S**
6. Run the completed Mininet Python Script
 - a. Open a terminal inside LAB1 folder by **right click – Open in Terminal**
 - b. `sudo python3 lab1.py`

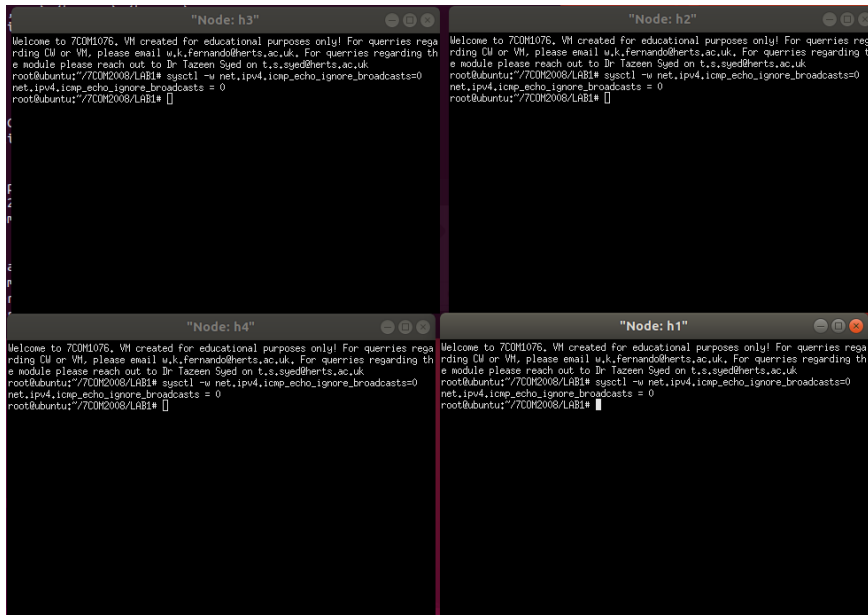
3. Network Transmission Modes

1. Try the following inside the Mininet Environment that you ran in the previous task (Introduction to Python API)
 - a. Unicast
 - i. `h1 ping h2`
Press **CTRL + C** to end transmission.
 - b. Broadcasting
 - i. `xterm h1 h2 h3 h4`

Inside each opened xterm window, type in the following.

ii. `sysctl -w net.ipv4.icmp_echo_ignore_broadcasts=0`

This command will disable the echo ignore setting for ICMP broadcasts on each host



iii. On h1 terminal type `ping -b 10.0.0.255`

You will see a broadcast response from h2, h3, and h4, arriving at h1.

iv. Press CTRL + C to end the transmission and close all the xterm windows.

v. Exit Mininet (type `exit`) and launch the environment again using `sudo python3 lab1.py`

c. Multicasting

Open a second terminal as you did in task 1.1 and type

i. `sudo wireshark`

and enter the password as 123 if prompted

ii. Double click on `s1-eth1` as this is the ethernet h1 is connected to the network. You can observe the ethernet connections by typing in the command **net** to the Mininet

On the Mininet, type in **xterm h1 h3 h4** and type the below

iii. `echo 0 > /proc/sys/net/ipv4/icmp_echo_ignore_broadcasts`

iv. `ip route add 224.0.0.0/8 dev h1-eth0`

Adjust for each host's interface (h3-eth0, h4-eth0, etc.)

On the h1 terminal, initiate a multicast ping

v. `ping 224.0.0.1`

vi. Observe the traffic in Wireshark